

Setup

The following code sets up the system by downloading the library from the web.

Make a closed figure function

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

Make a video of drawing a face

```
In [12]: # Playing with b

# Create a graph object with equal_axes)
plot = graph_funs(equal_axes=True, hide_axes=False)

# Setup the grid
# Determine the range for X based on reflections and translations.
# Determine the range for Y based on reflections and translations (look at maximum values).
plot.set_XY(minX=-10, maxX=+10, minY=0, maxY=+35)

# Draw different parts of the face and save in different frames

# Frame 1: draw the face:
make_closed_fig(plot, a=1.5, b=1.25, h=0, k=0, d=10, num_of_points=10)
plot.plot_funs(plot_title="Drawing the face", img_name="frame1.png")

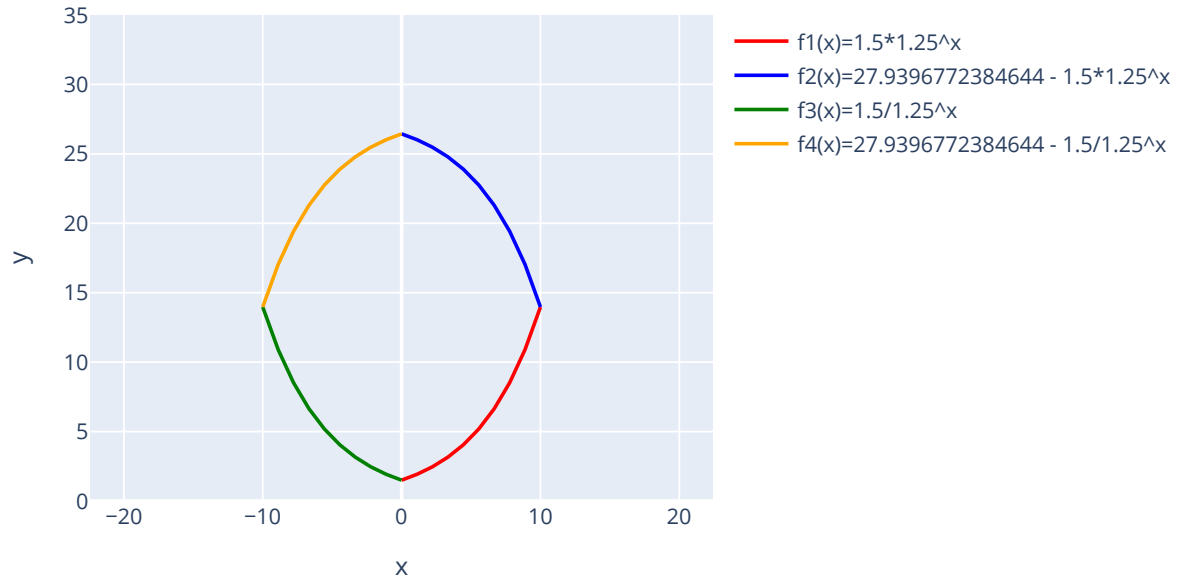
# Frame 2: Draw the eyes:
make_closed_fig(plot, a=.25, b=b, h=5, k=16, d=1.5, num_of_points=20)
make_closed_fig(plot, a=.25, b=b, h=-5, k=16, d=1.5, num_of_points=20)
plot.plot_funs(plot_title="Drawing the eyes", img_name="frame2.png")

# Frame 3: Draw the nose:
make_closed_fig(plot, a=.25, b=b, h=0, k=13, d=1, num_of_points=20)
plot.plot_funs(plot_title="Drawing the nose", img_name="frame3.png")

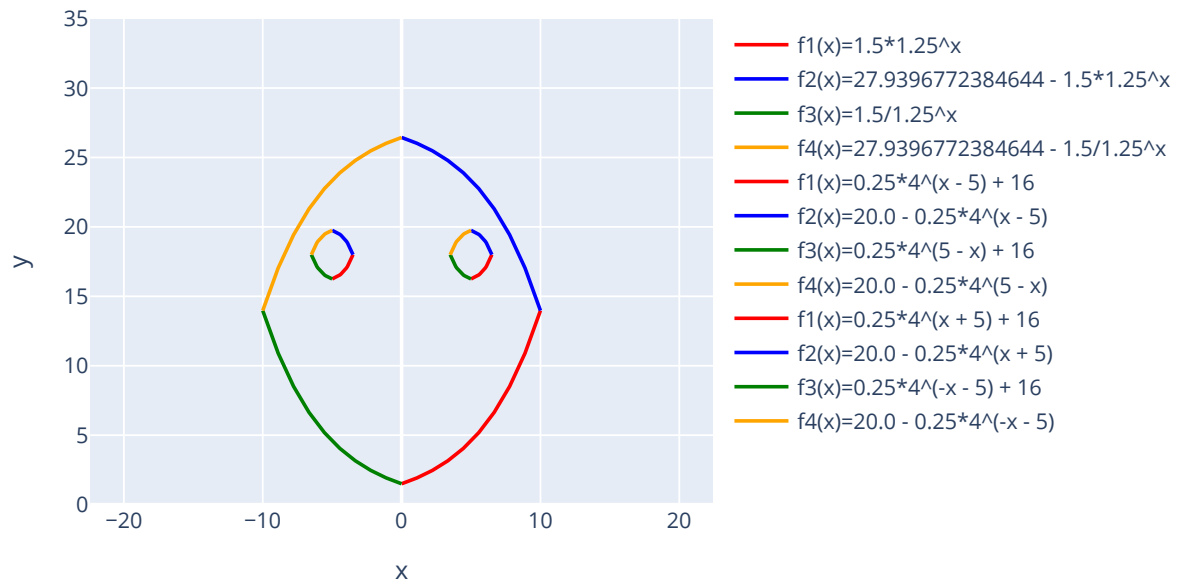
# Frame 4: draw the smile:
make_closed_fig(plot, a=1.5, b=1.25, h=0, k=6, d=1, num_of_points=10)
plot.plot_funs(plot_title="Drawing the mouth", img_name="frame4.png")

# Frame 5: Draw the horns:
make_closed_fig(plot, a=2.5, b=b, h=5, k=21, d=.75, num_of_points=20)
make_closed_fig(plot, a=2.5, b=b, h=-5, k=21, d=.75, num_of_points=20)
plot.plot_funs(plot_title="Drawing the horns", img_name="frame5.png")
```

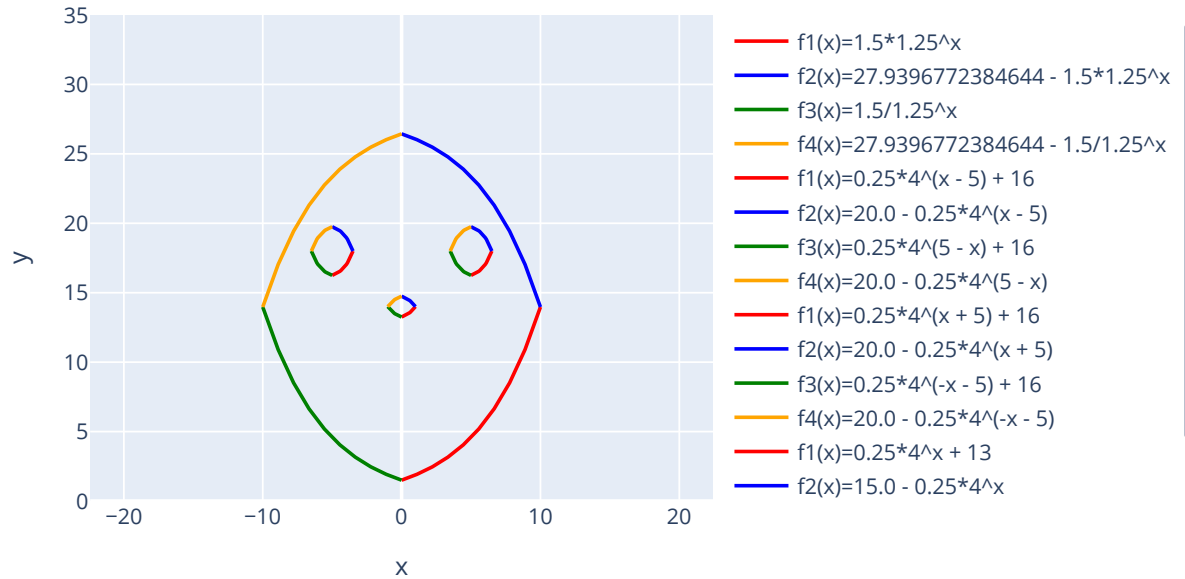
Drawing the face



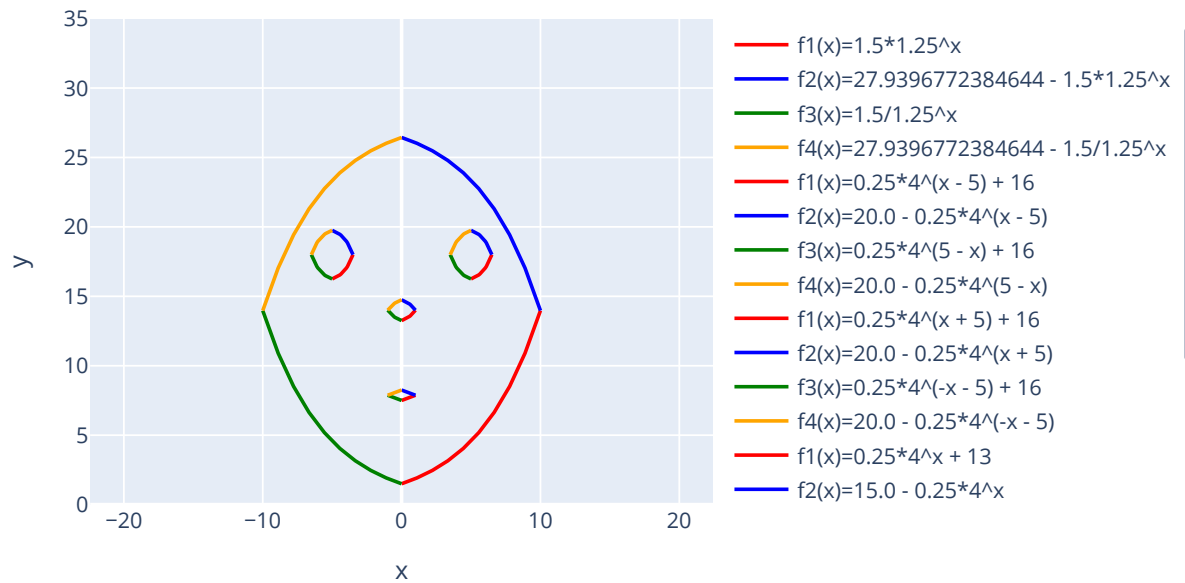
Drawing the eyes



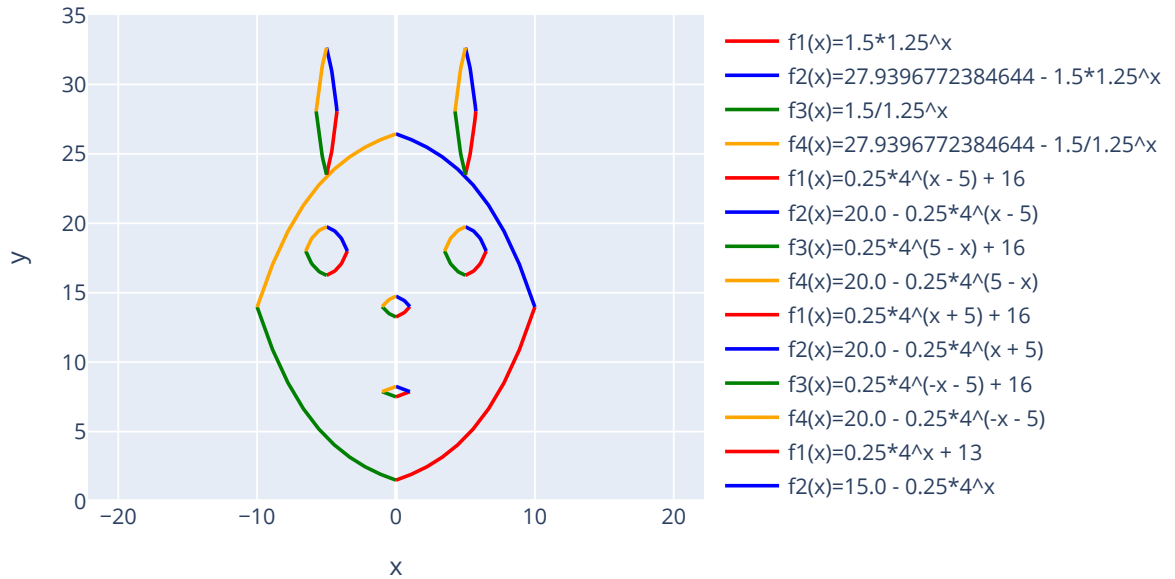
Drawing the nose



Drawing the mouth



Drawing the horns



In [13]: `!jupyter nbconvert --to html /content/drive/MyDrive/Colab_Notebooks/Untitled0.ipynb`

[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab_Notebooks/Untitled0.ipynb to html

[NbConvertApp] Writing 351848 bytes to /content/drive/MyDrive/Colab_Notebooks/Untitled0.html

In []: